Exponentials & Logarithms

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponent rules**

Simplify $x^5 \cdot x^3$.

$x^{5+3} = x^8$.

2 **[Foundational]** • **Target: Exponent rules**

Simplify $(2x^3)^4$.

$2^4 x^{12} = 16x^{12}$.

3 **[Foundational]** • **Target: Exponent rules**

Simplify $\frac{5^6}{5^2}$.

$5^{6-2} = 5^4 = 625$.

4 **[Foundational]** • **Target: Solving exponential equations**

Solve $3^x = 81$.

$81 = 3^4$, so $x = 4$.

5 **[Foundational]** • **Target: Solving exponential equations**

Solve $2^{x+3} = 64$.

$64 = 2^6$, so $x + 3 = 6$, giving $x = 3$.

6 **[Foundational]** • **Target: Solving exponential equations**

Solve $5^{2x} = 125$.

$125 = 5^3$, so $2x = 3$, giving $x = \frac{3}{2}$.

7 **[Foundational]** • **Target: Logarithm definition and evaluation**

Evaluate $\log_2 64$.

$2^6 = 64$, so $\log_2 64 = 6$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Logarithm definition and evaluation

Evaluate $\log_5 1$.

$5^0 = 1$, so $\log_5 1 = 0$.

9 [Intermediate] • Target: Logarithm definition and evaluation

Rewrite $\log_3 81 = 4$ in exponential form, and verify.

$3^4 = 81$, which checks out.

10 [Intermediate] • Target: Logarithm laws

Simplify $\log 8 + \log 5$ as a single logarithm.

$\log(8 \times 5) = \log 40$.

11 [Intermediate] • Target: Logarithm laws

Simplify $\log_2 48 - \log_2 3$.

$\log_2 \left(\frac{48}{3}\right) = \log_2 16 = 4$.

12 [Intermediate] • Target: Logarithm laws

Write $3\log_2 x + \log_2 y$ as a single logarithm.

$\log_2 (x^3 y)$.

13 [Intermediate] • Target: Solving logarithmic equations

Solve $\log_4 (x + 5) = 2$.

$4^2 = 16 = x + 5$, so $x = 11$.

14 [Intermediate] • Target: Solving logarithmic equations

Solve $\log_2 (x) + \log_2 (x - 2) = 3$.

$\log_2 [x(x - 2)] = 3$, so $x(x - 2) = 8$, giving $x^2 - 2x - 8 = 0$, i.e. $(x - 4)(x + 2) = 0$. So $x = 4$ or $x = -2$; but $x = -2$ makes the original logarithms undefined, so the only valid solution is $x = 4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 1.01 Heart of Algebra

A population of bacteria starts at 300 and doubles every 4 hours. Write an exponential model and find the population after 12 hours.

$P(t) = 300 \cdot 2^{t/4}$. $P(12) = 300 \cdot 2^3 = 300 \times 8 = 2400$.

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16 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 2.01 Quadratics & Polynomials

A car worth 24{,}000 dollars depreciates at 12 percent per year. Find its value after 3 years, to the nearest dollar.

$V = 24000(0.88)^3 = 24000 \times 0.681472 \approx 16{,}355$ dollars.

17 [Challenging] • Target: Graphs of exponential functions • Cross-Lesson: 3.01 Functions & Graphs

The graph of $y = 2^x$ is shown below. Use the graph to estimate the value of x for which $y = 8$, then confirm your answer algebraically.

From the graph, $x \approx 3$. Algebraically: $2^x = 8 = 2^3$, so $x = 3$.

18 [Challenging] • Target: Natural logarithms and e • Cross-Lesson: 4.01 Ratios, Rates & Data

Evaluate $\ln(e^5)$.

Since $\ln$ and e^x are inverse functions, $\ln(e^5) = 5$.

19 [Challenging] • Target: Natural logarithms and e • Cross-Lesson: 5.01 Statistics & Probability

Solve $e^x = 20$ for x, to the nearest hundredth, using natural logarithms.

$x = \ln(20) \approx 2.996 \approx 3.00$.

20 [Challenging] • Target: A well-designed word problem: comparing two investment accounts • Cross-Lesson: 6.01 Geometry

This question has three parts. An investment of 5000 dollars is placed in an account that grows according to $A(t) = 5000e^{0.04t}$, where t is measured in years. (a) Find the value of the investment after 10 years, to the nearest dollar. (b) Find how many years it takes for the investment to double, to the nearest tenth of a year, using natural logarithms. (c) A second account offers simple (non-compounded) growth of 5 percent per year on the same initial 5000 dollars. Determine which account has the greater value after 10 years, and by how much, to the nearest dollar.

(a) $A(10) = 5000e^{0.4} \approx 5000(1.4918) \approx 7459$ dollars. (b) Set $10000 = 5000e^{0.04t}$, so $2 = e^{0.04t}$, giving $\ln 2 = 0.04t$, so $t = \frac{\ln 2}{0.04} \approx \frac{0.6931}{0.04} \approx 17.3$ years. (c) Simple growth value: $5000 + 5000(0.05)(10) = 5000 + 2500 = 7500$ dollars. Since $7500 > 7459$, the simple-growth account is greater after 10 years, by about $7500 - 7459 = 41$ dollars.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exponent rules

Solve $6^x = 216$.

a. 3 ✓

b. 216

c. 4

d. 2

$216 = 6^3$, so $x = 3$.

22 [Foundational] • Target: Solving exponential equations

Solve $4^x = 16$.

a. 16

b. 2 ✓

c. 1

d. 3

$16 = 4^2$, so $x = 2$.

23 [Foundational] • Target: Logarithm definition and evaluation

Solve $5^x = 125$.

a. 3 ✓

b. 125

c. 4

d. 2

$125 = 5^3$, so $x = 3$.

24 [Foundational] • Target: Logarithm laws

Solve $3^x = 9$.

a. 1

b. 9

c. 2 ✓

d. 3

$9 = 3^2$, so $x = 2$.

25 [Foundational] • Target: Solving logarithmic equations

Solve $4^x = 64$.

a. 3 ✓

b. 4

c. 64

d. 2

$64 = 4^3$, so $x = 3$.

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26 [Foundational] • Target: Exponential growth and decay models

Solve $2^x = 4$.

a. 3

b. 4

c. 1

d. 2 ✓

$4 = 2^2$, so $x = 2$.

27 [Foundational] • Target: Graphs of exponential functions

Solve $9^x = 729$.

a. 3 ✓

b. 4

c. 2

d. 729

$729 = 9^3$, so $x = 3$.

28 [Foundational] • Target: Natural logarithms and e

Solve $\log_3(x + 3) = 4$.

a. 80

b. 76

c. 81

d. 78 ✓

$3^4 = 81 = x + 3$, so $x = 78$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: comparing two investment accounts

Solve $\log_4(x - 12) = 1$.

a. 18

b. 14

c. 4

d. 16 ✓

$4^1 = 4 = x - 12$, so $x = 16$.

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30 [Intermediate] • Target: Exponential Equations & Radicals

Solve $\log_4(x+5)=1$.

a. 1

b. -3

c. 4

d. -1 ✓

$4^1=4=x+5$, so $x=-1$.

31 [Intermediate] • Target: Logarithmic Properties & Evaluation

Solve $\log_5(x+9)=2$.

a. 16 ✓

b. 14

c. 25

d. 18

$5^2=25=x+9$, so $x=16$.

32 [Intermediate] • Target: Solving Logarithmic Equations

Solve $\log_3(x-6)=1$.

a. 11

b. 3

c. 9 ✓

d. 7

$3^1=3=x-6$, so $x=9$.

33 [Intermediate] • Target: Exponential Growth, Decay & Compound Interest

Solve $\log_5(x-13)=2$.

a. 25

b. 40

c. 36

d. 38 ✓

$5^2=25=x-13$, so $x=38$.

34 [Intermediate] • Target: Exponent rules

Solve $\log_4(x+13)=4$.

a. 256

b. 241

c. 245

d. 243 ✓

$4^4=256=x+13$, so $x=243$.

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35 [Intermediate] • Target: Solving exponential equations

A population of bacteria starts at 497 and doubles every 3 hours. Find the population after 12 hours.

a. 1988

b. 3976

c. 7952 ✓

d. 8449

$P(t) = 497 \cdot 2^{t/3}$. $P(12) = 497 \cdot 2^4 = 497 \times 16 = 7952$.

36 [Intermediate] • Target: Logarithm definition and evaluation

A population of bacteria starts at 142 and doubles every 2 hours. Find the population after 4 hours.

a. 710

b. 568 ✓

c. 284

d. 852

$P(t) = 142 \cdot 2^{t/2}$. $P(4) = 142 \cdot 2^2 = 142 \times 4 = 568$.

37 [Intermediate] • Target: Logarithm laws

Solve $e^x = 898$ for x, to the nearest hundredth, using natural logarithms.

a. 6.80 ✓

b. 6.30

c. 7.80

d. 7.30

$x = \ln(898) \approx 6.80$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 4.01 Ratios, Rates & Data

Solve $e^x = 468$ for x, to the nearest hundredth, using natural logarithms.

a. 5.65

b. 6.65

c. 6.15 ✓

d. 7.15

$x = \ln(468) \approx 6.15$.

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39 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 5.01 Statistics & Probability

Simplify $\log_3 567 - \log_3 7$.

a. 4 ✓

b. 3

c. 560

d. 5

$\log_3\left(\frac{567}{7}\right) = \log_3 81 = 4$.

40 [Challenging] • Target: Graphs of exponential functions • Cross-Lesson: 6.01 Geometry

Simplify $\log_2 112 - \log_2 7$.

a. 5

b. 105

c. 3

d. 4 ✓

$\log_2\left(\frac{112}{7}\right) = \log_2 16 = 4$.

41 [Challenging] • Target: Exponential Equations & Radicals • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Solve the exponential equation for x: $27^{x-1} = 9^{x+2}$.

a. 5

b. 6

c. 7 ✓

d. 8

Convert to base 3: $(3^3)^{x-1} = (3^2)^{x+2} \Rightarrow 3^{3x-3} = 3^{2x+4}$. Equating exponents: $3x - 3 = 2x + 4 \Rightarrow x = 7$.

42 [Challenging] • Target: Exponential Equations & Radicals • Cross-Lesson: 1.01 Heart of Algebra

If $2^x = 5$, what is the value of 2^{x+3}?

a. 8

b. 15

c. 40 ✓

d. 125

$2^{x+3} = 2^x \times 2^3 = 5 \times 8 = 40$.

43 [Challenging] • Target: Logarithmic Properties & Evaluation • Cross-Lesson: 2.01 Quadratics & Polynomials

If $\log_{10}(2) \approx 0.301$, what is the approximate value of $\log_{10}(20)$?

a. 1.301 ✓

b. 2.301

c. 0.602

d. 3.010

$\log_{10}(20) = \log_{10}(2 \times 10) = \log_{10}(2) + \log_{10}(10) = 0.301 + 1 = 1.301$.

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44 [Challenging] • Target: Solving Logarithmic Equations • Cross-Lesson: 3.01 Functions & Graphs

Solve for x: $\log(x) + \log(x-3) = 1$ in base 10.

a. 5 ✓

b. 2

c. -2

d. 5 and -2

$\log(x(x-3)) = 1 \Rightarrow x^2 - 3x = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x-5)(x+2) = 0$. Since $x > 3$, the only valid solution is $x = 5$.

45 [Challenging] • Target: Exponential Growth, Decay & Compound Interest • Cross-Lesson: 4.01 Ratios, Rates & Data

A radioactive isotope decays according to $N(t) = 800(0.5)^{t/12}$, where t is measured in days. How much of the isotope remains after 36 days?

a. 50 g

b. 100 **g** ✓

c. 200 g

d. 400 g

At $t = 36$: $t/12 = 36/12 = 3$ half-lives. $N(36) = 800(0.5)^3 = 800 \times \frac{1}{8} = 100$ grams.

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